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A survey and analysis of traditional medicinal plants as used by the Zulu, Xhosa and Sotho

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ABSTRACT

A coded list of 794 traditional Zulu medicinal plants is presented with a key to the ailments concerned. Xhosa and Sotho usage of these plants is incorporated. Medicinal usage in the pteridophytes, gymnosperms, monocotyledons and dicotyledons is tabulated. Ailments are categorized and discussed with an analysis of the plant families involved in their treatment. Patterns of usage between related plants are observed and some potentially effective or dangerous characteristic family constituents are briefly outlined.

UITTREKSEL

'n Gekodeerde lys van 794 tradisionele medisinale plante van die Zoeloes word aangebied met 'n sleutel tot die betrokke ongesteldhede. Die gebruik van hierdie plante deur Xhosas en Sotho's word ook vermeld. Benutting van die pteridofiete, gimnosperms, monokotiele en dikotiele is getabuleer. Ongesteldhede word gekategoriseer en bespreek met 'n ontleding van die plantfamilies wat by die behandeling betrokke is. Benuttingspatrone tussen verwante plante is waargeneem en sekere potensieel doeltreffende of gevaarlike stowwe wat kenmerkend in families voorkom, word kortliks aangetoon.

INTRODUCTION

The data on which this paper is based come mainly from a literature survey conducted by the author towards the compilation of a Pharmacapoeia of Zulu Folk Medicine, which was initiated at the University of Zululand by Dr A. H. Scott in 1986. Xhosa and Sotho usage of the plants, which is closely allied, has been included. Data from the author's previous fieldwork in Transkei and limited personal interviews with Zulu and Xhosa traditional healers is included and further data have been abstracted from a list of the Botanical Research Institute's holdings of medicinal plants. The main sources of information on usage have been 1, A. T. Bryant, whose work on Zulu medicine was originally published in the *Annals of the Natal Museum* in 1909 but later destroyed by fire and republished in 1966 (Bryant 1966); 2, J. Gerstner, who published his work on Zulu plant names and usage between 1938 and 1941 (Gerstner 1938, 1939, 1941); and 3, Mairn Hulme, who included Zulu usage and also illustrated her book (Hulme 1954). Watt & Breyer-Brandwijk (1962) was also consulted and additional information on Sotho usage is from A. Jacot Guillarmod (Jacot Guillarmod 1971).

It would be of great interest to compare the number of plant species used with the number of plant species known to be available, but research in this field, embracing both ecological and trading aspects, is beyond the scope of this paper. Comprehensive research into the toxicity and active chemical principles in the plants used is also beyond the scope of the paper.

DISCUSSION OF LIST OF MEDICINAL PLANTS

A total number of 794 plant species with specific usage known has been recorded. These are grouped as pteridopytes, gymnosperms, monocotyledons and dicotyledons. They are arranged in alphabetic order according to family and within the family alphabetically according to genus and species. All plant names used in the literature surveyed have been recorded and the names listed have been updated according to Gibbs Russell *et al.* (1985, 1987). Authors' names have been omitted from the coded list to save space so that uses could be tabulated and similarities in the usage of related species could be easily seen.

Plants of the same genus which could not be identified to species level have been counted as one species unless, as in the case of the *Plectranthus* spp. illustrated by Hulme (1954), it is quite evident that different species are being referred to.

The species listed have been coded according to ailment. These ailments have been very broadly categorized and determined largely by symptom, as is traditional treatment. Overlaps occur, as in the case of bladder and urethral disorders, which have been classed by the author as a common renal ailment but could be caused or affected by procreation-related ailments such as sterility and venereal disease.

The presence of a known or suspected toxic element recorded in the literature surveyed is indicated by an asterisk in the coded list at the end of this article. The toxic element may not always be known in the part of the plant used.

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Key to ailments:

A — Sexual
B — Sterility
C — Venereal
D — Gynaecological
E — Newly born infants
F — Gastro-intestinal
G — Renal
H — Tonic
I — Growths
J — Respiratory
K — Febrile
L — Headaches
M — Heart
N — Nervous
O — Pain producing
P — Leprosy
Q — Dental
R — Sorcery
S — Snake-bite
T — Skin
U — Sprains, fractures
V — Eyes, ears and nose
W — Insecticidal
X — Charm
Y — Animal

a — Includes use as anthelmintic
c — Includes enema administration
d — Includes use against diarrhoea
e — Includes use as an emetic
i — Given to infants or children
p — Taken regularly during pregnancy
r — Rubbed into scarifications
s — Snuffed or inhaled
* — Toxic

ANALYSIS

In this analysis of ailments and methods of treatment, families, of which three or more species are used for the same ailment, are highlighted.

Families marked with an asterisk are those of which one or more species used for the specific ailment have been recorded as toxic.

Procreation-related ailments (A–E in key to ailments and coded list of plants)

A. *Sexual* remedies include aphrodisiacs and a large number of love charm emetics, which are normally taken by men. Some medicines are administered as charms to secure the fidelity of the beloved or to harm a rival in cases of suspected infidelity or to protect the user against the effects of such medicine.

Of the 109 species recorded as sexual remedies, three or more occur in the following 12 families, accounting for 79:

13 Orchidaceae
13 Fabaceae*
11 Liliaceae*
7 Rubiaceae
5 Euphorbiaceae*
4 Apiaceae
4 Acanthaceae
4 Gentianaceae
4 Scrophulariaceae
4 Acanthaceae
4 Asteraceae*
3 Amaryllidaceae*
3 Caryophyllaceae

95 of the 109 species are used as love charm emetics.

B. *Sterility* remedies include cures for both women and men. Barrenness in women is usually treated with a purgative or an enema or medicine may be directly introduced into the womb. Sterility and impotence in men is treated with orally taken infusions, or powdered medicines may be blown through the urethra (Bryant 1966). Also included are medicines taken, usually orally, by a couple desiring a particular gender in a child, or medicines taken by both parents after a miscarriage.

Of the 45 species recorded as sterility remedies, three or more occur in the following six families, accounting for 25:

7 Orchidaceae
4 Iridaceae*
5 Fabaceae
3 Liliaceae*
3 Amaryllidaceae
3 Rubiaceae

C. *Venereal diseases* (syphilis and gonorrhoea) may be treated with orally administered decoctions or, in cases of discharge, medicine may be inserted directly into the penis or vagina in liquid form or as a pill of pounded leaves, and charred powdered leaves or a poultice may be applied directly to venereal sores (Bryant 1966).

Of the 34 species recorded as venereal disease remedies, three or more occur in the following five families, accounting for 19:

6 Liliaceae*
4 Asteraceae
3 Ranunculaceae*
3 Fabaceae*
3 Solanaceae*

D. *Gynaecological* remedies include medicines taken regularly during pregnancy to ensure a safe delivery and a healthy child. These may be mixtures of various ingredients, usually roots (Gerstner 1941) or may be made from the roots of a single plant, soaked in water, which is drunk daily by the expectant mother. Also included are medicines administered during childbirth, or for painful or delayed menstruation, to stimulate breast development or the flow of milk, or to procure abortions.

Of the 78 species recorded, three or more occur in the following six families, accounting for 37:

12 Liliaceae*
10 Asteraceae*
4 Amaryllidaceae*
4 Ebenaceae*
4 Scrophulariaceae
3 Euphorbiaceae*

Of the 78 species used for gynaecological purposes, 38 are recorded as taken regularly during pregnancy, three or more of which occur in the following three families, accounting for 16:

8 Liliaceae*
5 Asteraceae*
3 Amaryllidaceae*

E. *Newly-born infants* are sometimes given purification purges at birth and later at weaning; these may be administered orally or as an enema. Medicines may be applied to the navel or fontanel at birth and others may be applied to the mother's breast at weaning. Sometimes newly born infants are held in the smoke of a burning plant in a protective or purification ritual.

Of the 18 species recorded, seven occur in Liliaceae.

Stress-related ailments (L–O in key and coded list of plants)

(*Note*—the term stress-related is used here to refer to ailments that are caused by psychological stress, although there may be other causes.)

L. *Headaches* are considered by traditional healers to be caused by emotional or mental disturbance. They are often treated by snuffed or inhaled medicines and occasionally leaves are wrapped around the head.

Of the 44 species recorded, three or more come from the following five families, acounting for 22:

6 Euphorbiaceae*
6 Asteraceae*
4 Ranunculaceae*
3 Fabaceae*
3 Asclepiadaceae*

M. *Cardiac* ailments include high blood pressure, chest pain not associated with respiratory complaints and bad dreams believed to be caused by heart problems (Bryant

1966). Treatment may be administered in the form of an emetic or enema.

Of the 21 species recorded, three occur in the following two families, accounting for six:

3 Liliaceae* 3 Fabaceae*

N. *Nervous* or psychological ailments include hysteria, mental disturbance, nightmares, states of believed bewitchment, states of impurification after the death of a kraal member and states of trance which need to be induced in the diviner to enable her to fulfil her function. Emetics and snuffed or inhaled medicines are frequently used for these purposes.

Of the 133 species recorded for nervous ailments, three or more come from the following 12 families, accounting for 81:

16 Fabaceae*
12 Asteraceae*
8 Liliaceae*
8 Rubiaceae
7 Orchidaceae
8 Asclepiadaceae*
5 Iridaceae*
4 Euphorbiaceae*
4 Apiaceae
3 Hypoxidaceae*
3 Sapindaceae
3 Solanaceae*

O. *Pain producing* ailments include pain in the joints or back, rheumatism and also paralysis. Although these may obviously not always have a psychological cause, there is a traditional belief that the joints are the most vulnerable area to the entrance of evil spirits (Ngubane 1977) and conversion disorders often do take the form of pain in the joints and sometimes paralysis. Dried powdered medicine is frequently rubbed into freshly cut scarifications to treat these conditions and other forms of administration such as enemas and emetics are also used.

Of the 46 species recorded, three or more come from the following families, accounting for 13:

7 Liliaceae* 3 Meliaceae*
3 Fabaceae*

Miscellaneous ailments (F–K and Q–Y in key and coded list of plants)

F. *Gastro-intestinal* ailments include:

1, stomach ache and constipation, which are treated with enemas or orally administered purges;

2, intestinal worms which are treated with orally administered anthelmintics or enemas;

3, diarrhoea and dysentery which are treated with an orally administered medicine, frequently followed by an enema of the same medicine;

4, nausea or the accumulation of bile is treated with an emetic;

5, indigestion, for which small pieces of root, bark or leaves may be chewed;

6, haemorrhoids and a condition which Bryant (1966) and Ngubane (1977) refer to as gangrenous rectitis, which is frequently treated with an enema, or a locally applied lotion or powder and may also be treated with an orally taken decoction.

Of the 318 species recorded for gastro-intestinal complaints, three or more occur in the following 35 families, accounting for 252:

40 Asteraceae*
27 Liliaceae*
19 Fabaceae*
12 Rubiaceae
13 Euphorbiaceae*
11 Iridaceae*
8 Lamiaceae
7 Crassulaceae*
7 Asclepiadaceae*
7 Convolvulaceae
7 Cucurbitaceae*
6 Ebenaceae*
6 Celastraceae*
5 Apiaceae
5 Myrsinaceae*
5 Verbenaceae
5 Solanaceae*
4 Poaceae
4 Cyperaceae
4 Amaryllidaceae*
4 Polygonaceae
4 Amaranthaceae
4 Mesembryanthemaceae
4 Rosaceae
4 Geraniaceae
4 Meliaceae*
4 Anacardiaceae*
4 Scrophulariaceae
3 Aspidiaceae*
3 Ranunculaceae
3 Rutaceae
3 Sapindaceae*
3 Gentianaceae
3 Apocynaceae*

Of the 318 species recorded as gastro-intestinal remedies 46 are recorded as anthelmintics and 54 as diarrhoea and dysentery remedies. These species may also be used for other gastro-intestinal purposes.

Of the 46 recorded anthelmintics, three or more occur in the following four families, accounting for 22:

8 Asteraceae* 4 Myrsinaceae*
7 Liliaceae* 3 Aspidiaceae*

Of the 54 recorded diarrhoea and dysentery remedies, three or more occur in the following six families, accounting for 30:

10 Fabaceae* 3 Sapindaceae*
7 Iridaceae* 3 Rubiaceae
4 Geraniaceae 3 Asteraceae

G. *Renal* ailments include kidney and urinary tract complaints. Medicines may be orally administered, sometimes followed by an enema of the same medicine after three days of treatment, or medicines may be rubbed into incisions cut in the loins or inserted directly into the urethra or a poultice may be applied externally.

Of the 44 species recorded for renal ailments, three or more are found in the following five families, accounting for 19:

5 Amaryllidaceae* 3 Apocynaceae*
5 Asteraceae* 3 Euphorbiaceae*
3 Liliaceae*

H. *Debility* ailments include general weakness, a low resistance to infections, blood impurities and a lack of appetite. They are treated with orally taken tonics, emetics or enemas.

Of the 58 species recorded for debility, three or more come from the following six families, accounting for 24:

8 Asteraceae 3 Apocynaceae*
4 Euphorbiaceae* 3 Periplocaceae
3 Fabaceae 3 Rubiaceae

I. *Swellings or growths* include swollen glands, scrofulous and dropsical swellings and external or internal swellings or lumps which may or may not be cancerous. Medicines may be administered orally, sometimes followed by a poultice made from the same ingredients, or powdered medicines may be rubbed into incisions cut around the swelling.

Of the 31 species used to treat swellings, three or more come from the following two families, accounting for eight:

5 Euphorbiaceae* 3 Asteraceae*

J. *Respiratory* ailments include chest pain from pleurisy or bronchitis, coughs, sore throats and asthma or catarrh. They are frequently treated with emetics, which perform an expectorant function, or medicines may be chewed, drunk, snuffed, inhaled or rubbed into incisions on the chest.

Of the 144 species recorded for respiratory ailments, three or more come from the following 14 families, accounting for 92:

27 Asteraceae*
12 Fabaceae*
8 Euphorbiaceae*
6 Liliaceae*
5 Amaryllidaceae*
5 Apiaceae*
5 Lamiaceae
5 Verbenaceae
4 Rubiaceae
3 Phytolaccaceae*
3 Brassicaceae
3 Capparaceae*
3 Ebenaceae*
3 Celastraceae*

K. *Febrile* complaints include influenza, colds and fevers, including malaria and rheumatic fever. They are frequently treated with emetics. Snuffed or inhaled medicines may also be administered and the patient may be steamed or bathed to reduce the fever.

Of the 123 species involved, three or more come from the following 13 families, accounting for 82:

21 Asteraceae*
12 Fabaceae
10 Lamiaceae
6 Liliaceae*
6 Euphorbiaceae*
5 Rubiaceae
4 Amaryllidaceae*
3 Iridaceae*
3 Rutaceae
3 Amaranthaceae
3 Apiaceae
3 Scrophulariaceae
3 Lobeliaceae

P. *Leprosy* is usually now referred by traditional healers to the hospitals. Only two remedies have been recorded and they are from Melianthaceae and Passifloraceae.

Q. *Toothache* and sore gums are treated with lotions or powders rubbed on to the painful area.

Of the 30 species recorded for toothache, three or more are from the following three families, accounting for 12:

5 Solanaceae*
4 Asteraceae*
3 Euphorbiaceae*

R. *Sorcery* is believed to be the cause of many illnesses and certain plants are believed to be used by sorcerers to bring about evil. The same plants may be taken as an antidote to the disease thus brought about, usually in the form of an emetic.

Of the 16 species recorded as sorcerer's medicines, three each are from the following two families, accounting for six:

3 Euphorbiaceae*
3 Apiaceae*

S. *Snake bite* remedies may be taken or locally applied, the same medicine often being administered at the same time in both forms. Dried ground snakes' heads are sometimes an ingredient in the medicine.

Of the 43 species recorded, three or more come from the following six families, accounting for 20:

4 Thymelaeaceae*
4 Asteraceae*
3 Phytolaccaceae*
3 Fabaceae*
3 Euphorbiaceae*
3 Apocynaceae*

T. *Skin* complaints include sores, wounds, burns and rashes. These complaints may be treated by applied lotions, poultices or washes. Washes may also be given to reduce the temperature in fevers. Also included are hair restorers.

Of the 100 species recorded, three or more come from the following 10 families, accounting for 64:

15 Asteraceae*
11 Fabaceae*
9 Solanaceae*
7 Euphorbiaceae*
4 Liliaceae*
4 Amaryllidaceae*
4 Thymelaeaceae
4 Acanthaceae
3 Verbenaceae
3 Lamiaceae

U. *Fractures, sprains and bruises* may be treated with lotions, or powder may be rubbed into scarifications as an anti-inflammant around the site of the damage.

Of the 40 species recorded, three or more come from the following three families, accounting for 12:

6 Asteraceae*
3 Euphorbiaceae*
3 Crassulaceae*

V. *Ear, eye and nose* complaints are treated with lotions or sap directly squeezed from the plants.

Of the 39 species recorded, three or more come from the following four families, accounting for 19:

7 Liliaceae*
6 Asteraceae
3 Crassulaceae*
3 Rosaceae

W. *Insecticides and piscicides* include medicines applied to rid the body of lice and fleas and those used to repel or kill flies and those used to stun fish so that they may be caught.

Of the 21 species recorded, three or more come from the following two families, accounting for 12:

9 Fabaceae*
3 Asteraceae

X. *Charms* may be applied to placate evil spirits, for protection against enemies and storms, for good luck or to make the user liked. They may be bathed with, worn, sprinkled or burnt in the home or fields or cultivated as protective plants.

Of the 182 species recorded, three or more come from the following 14 families, accounting for 126:

27 Liliaceae*
18 Orchidaceae
15 Asclepiadaceae*
11 Asteraceae*
9 Amaryllidaceae*
9 Crassulaceae*
8 Iridaceae*
7 Fabaceae*
7 Euphorbiaceae*
6 Apiaceae
3 Rhamnaceae
3 Scrophulariaceae
3 Rubiaceae

Y. *Animals* may be given medicines to prevent or cure disease. Also included are medicines applied as theft deterrents, either in the field or on eggs or drying animal skins.

Of the 102 species used to treat or deter animals, three or more come from the following 11 families, accounting for 53:

11 Asteraceae*
9 Fabaceae*
7 Liliaceae*
6 Asclepiadaceae*
5 Lamiaceae
3 Amaryllidaceae*
3 Dioscoreaceae*
3 Urticaceae
3 Geraniaceae
3 Vitaceae*

It may be observed (Table 1) that among the medicinally used plants recorded, a higher proportion of monocotyledons are used as externally applied charms and for procreation-related purposes and that there is a slightly wider range of medicinal usage among the dicotyledons. Further analysis reveals that among the mono-

cotyledons 43 % of the species are used for procreation-related ailments, 37 % as externally applied charms and 26 % for possibly stress-related ailments whereas among the dicotyledons 27 % are used for procreation-related ailments, 26 % for possibly stress-related ailments and 19 % as externally applied charms. In both groups the greatest proportion of plants used for other ailments are used for gastro-intestinal purposes (32 % of monocotyledons and 42 % of dicotyledons) and respiratory ailments (11 % of monocotyledons and 20 % of dicotyledons). This is reflected in Tables 2 & 3.

DISCUSSION

Perception of aetiology also determines treatment. According to Bryant (1966) and Ngubane (1977), most common ailments are believed to be caused by an excess of bile or gall, which needs to be removed. Diseases believed to be caused by evil spirits or pollution also require catharsis. This explains the wide use of emetics, enemas and purgatives. A total of 238 of the plants recorded for this study, i.e. 30 %, are used as emetics—where use as an emetic has been recorded with no specific ailment, the plant has been considered to be used for gastro-intestinal purposes. Emetics are not normally given to young children but enemas are and are considered by medical staff to be the cause of some of the poisoning cases and liver damage seen in hospitals (Savage & Hutchings 1987). Forms of administration are not always recorded in the sources consulted and more plants are probably used for enemas than appear on the list. Species indicated as being used for children are frequently administered in this way. Patterns of usage between closely related species, such as the use of pteridophytes as anthelmintics, Orchidaceae as love charm emetics, Ranunculaceae species for headaches, Thymelaeaceae species for skin complaints and snake bite or Lamiaceae for febrile conditions and various Asclepiadaceae for nervous complaints, are discernible in the list.

Such patterns of usage could obviously indicate that related plants share chemical constituents, which would in turn account for their possible effectiveness and/or toxicity. Some characteristic family constituents with some of their likely effects and potential dangers are given below.

Widespread steroidal saponins, cardiac glycosides in some of the Liliaceae and toxic alkaloids in the Amaryllidaceae are among the potentially dangerous consituents found in monocotyledonous plants, of which so many are used in the procreation-related ailments. Steroidal saponins may affect the sex hormones and are relatively harmless when taken by mouth but found highly toxic if they enter the blood stream (Trease & Evans 1983). This may happen if there is any damage to the mucous lining of the gastro-intestinal tract when enemas are administered as, according to Ngubane (1977), the dosage is controlled to enable the patient to retain the medicine for a time. This custom would also make enemas made with other toxic material more dangerous than emetics as absorption of material through the rectum is easier than through the small intestine. Steroid or triterpenoid saponins (which are common in the dicotyledons) are often found in the plants used as emetics and have the property of foaming and also frequently irritate the mucosa. This may account for their expectorant and decongestive action when used for chest ailments. Anti-microbial, cytostatic and anti-inflammatory activity have been demonstrated in saponins (Lower 1985).

Cardiotonics, which can have a diuretic action by increasing the renal bloodflow, may be found in members of various other families apart from Liliaceae and these include Apocynaceae and Asclepiadaceae, Rubiaceae and Solanaceae (see Oliver-Bever 1986). The diuretic action would be helpful in cases of gonorrhoea and also in various kidney or heart diseases which cause dropsical swellings. Various species of the above-mentioned families are used for venereal diseases, renal complaints, dropsical swellings or heart complaints and may be found effective.

The toxic Amaryllidaceae alkaloids produce gastro-intestinal upset (Jaspherson–Schib 1970). Highly toxic species are found in other alkaloid-containing families such as Apocynaceae, Solanaceae and Euphorbiaceae. These species also owe their effectiveness as purges and their potentially dangerous properties to extreme gastric irritation.

Antipyretic, protozoicidal and local anaesthetic properties are to be found in many of the West African species of the alkaloid-rich Rubiaceae family (Oliver-Bever 1986) and members of the family locally used for febrile ailments would probably merit further investigation. Alkaloids have a marked action on the central nervous system and can act as depressants (e.g. the sedative reserpine from *Rauvolfia vomitoria* Afzel.) or stimulants (e.g. the *Strychnos* alkaloids) which may account for the use of various species from families such as Euphorbiaceae and Solanaceae for nervous complaints.

Tannin has frequently been observed in the parts of the plant used in the treatment of dysentery and diarrhoea or for respiratory ailments and is a characteristic constituent of many of the families thus used (e.g. Rosaceae, Fabaceae, Geraniaceae). It is likely to be effective on account of its protein-precipitating properties which, in small doses, would form a protective, impermeable layer and also tend to prevent the development of bacteria—large doses would irritate the mucosa (Flück 1976).

The presence of volatile oils with possible carminative or antispasmodic activity is likely to account for the use of the closely related Lamiaceae and Verbenaceae and other aromatic families for coughs, colds, influenza and digestive disorders.

The snake bite cures are interesting. Many are known to be either toxic or else closely related to known toxic species, notably from families where cardioactive toxins (Melianthaceae, Loganiaceae, Apocynaceae and Asclepiadaceae) or alkaloids (Phytolaccaceae, Solanaceae and Asteraceae) are known or else from families where cytotoxic activity has been observed (Euphorbiaceae and Thymelaeaceae).

More fieldwork will undoubtedly reveal new usage of plants as the practice of herbal medicine is still very much alive. Patterns obviously exist in the usage of plants, but the perception thereof is influenced by the way in which the data are analysed and on the cultural

TABLE 1.—Medicinal usage in pteridophytes, gymnosperms, monocotyledons and dicotyledons

Group	Gen.	Spp.	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
Pterid.	7	9		1		3		6				2	3	1		1			1			1					2
Gymnosp.	2	3	1					1						1												2	
Monocot.	79	173	30	21	7	25	10	56	9	2	3	19	17	5	5	30	9				1	12	6	8	1	65	17
Dicot.	345	609	78	23	28	50	12	255	35	56	28	123	103	37	16	102	37	2	31	16	43	87	34	30	20	115	83
Total	433	794	109	45	35	78	22	318	44	58	31	144	123	44	21	133	46	2	32	16	44	100	40	38	21	182	102

A–Y: see key to list of medicinal plants.

TABLE 2.—Monocotyledons: families with more than 10 medicinally used species

	No. spp.	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
Liliaceae	60	11	3	7	12	7	27	3		1	6	6	2	3	9	7			1		4	1	7	1	27	7
Amaryllidaceae	22	3	3		4		4	5		2	5	4	2		2	1				1	4	1			9	3
Iridaceae	25	1	4		2	3	11				2	3			5							2			8	2
Orchidaceae	34	13	7		2		2								7	1									18	
Total	141	28	17	7	20	10	44	8		3	13	13	4	3	23	9			1	1	8	4	7	1	62	12

A–Y: see key to list of medicinal plants.

TABLE 3.—Dicotyledons: families with more than 10 medicinally used species

	No. spp.	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
Crassulaceae	14	1		1	1		7					1			1			1			1	3	3		9	
Fabaceae	74	13	5	3			19	2	3		12	12	3	3	16	3		1	1	3	11	1	2	10	7	9
Euphorbiaceae	31	5			3	1	13	3	4	5	8	6	6		4	2		3	3	3	7	3	2	1	7	5
Apiaceae	11	4					5			1	5	3	1		4				3						6	1
Asclepiadaceae	28	1	1		2	2	7		1		1		3	1	8	2				1		2	1		15	6
Verbenaceae	10				1		5	2		2	5	2			2	1				2	3		2			1
Lamiaceae	21	2				1	8	1	2		5	10	2					2		2	3			1	1	5
Solanaceae	15		1	3			5	1		2	2	1	2		3	2		5		2	9	2	1			2
Scrophulariaceae	12	4		1	4		4				1	3			2	1					2	1	1		4	
Acanthaceae	12	4				1	1		1		1				2			2		2	4		2		2	1
Rubiaceae	23	7	3	1	2	1	12		3	1	4	5		2	8	1		1		1	2	1			3	
Asteraceae	80	4	2	3	10		40	4	8	3	27	21	6	2	12	2		4	2	4	15	6	7	3	10	11
Total	331	45	12	12	23	6	126	13	22	14	71	64	23	8	62	14		19	9	20	57	19	21	14	64	41

A–Y: see key to list of medicinal plants.

perceptions of the researcher. Nevertheless, plant usage is patterned, and these patterns deserve further investigation.

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LIST OF ZULU MEDICINAL PLANTS

Key to ailments:

A - Sexual
B - Sterility
C - Venereal
D - Gynaecological
E - Newly born infants
F - Gastro-intestinal
G - Renal
H - Tonic
I - Growths
J - Respiratory
K - Febrile
L - Headaches
M - Heart
N - Nervous
O - Pain producing
P - Leprosy
Q - Dental
R - Sorcery
S - Snake-bite
T - Skin
U - Sprains, fractures
V - Eyes, ears and nose
W - Insecticidal
X - Charm
Y - Animal

a - Includes use as anthelmintic
c - Includes enema administration
d - Includes use against diarrhoea
e - Includes use as an emetic
i - Given to infants or children
p - Taken regularly during pregnancy
r - Rubbed into scarifications
s - Snuffed or inhaled
* - Toxic

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
PTERIDOPHYTA																									
ADIANTACEAE																									
Adiantum capillus-veneris				D						J	K	Ls													
Cheilanthes hirta						Fa				J	K														
Cheilanthes viridis																				T					
ASPIDIACEAE																									
Dryopteris athamantica*						Fa																			Y
Dryopteris inaequalis						Fa								Ns											
Polystichum sp.						Fa																			Y
CYATHEACEAE																									
Cythea dregei				Dp																					
DENNSTAEDTIACEAE																									
Pteridium aquilinum*				D		Fa																			
EQUISETACEAE																									
Equisetum ramosissimum*		B				Fi				K					Q									X	
GYMNOSPERMAE																									
STANGERIACEAE																									
Stangeria eriopus						Fe						Ls												X	
ZAMIACEAE																									
Encephalartos altensteinii*	Ae																								
Encephalartos villosus*																								X	
ANGIOSPERMAE - MONOCOTYLEDONAE																									
AMARYLLIDACEAE																									
Annocharis coranica*														N											Y
Apodolirion sp.						Fc																			
Boophane disticha*							G			Jec	Lec	N	Oec			T									
Brunsvigia spp.*		B		D		Fi	6c			J	K					T								X	
Clivia miniata*		B		Dp			G				K								S	T					
Clivia nobilis*																								X	
Clivia sp.		Ae																						X	
Crinum bulbispermum				Dp							K													X	
Crinum delagoense																									Y
Crinum macowanii																				T					
Crinum moorei																									Y
Crinum sp.							G	H	I	K															
Cyrtanthus breviflorus		Ae																						X	
Cyrtanthus contractus																								X	
Cyrtanthus mackenii																								X	
Cyrtanthus obliquus		B					Gc		I	J		Ls									Ur				
Cyrtanthus stenanthus																								X	
Cyrtanthus tuckii																								X	
Haemanthus albiflos								J																	
Haemanthus coccineus*						Fe																			
Scadoxus multiflorus*		Ae																							
Scadoxus puniceus*	Dp	Fe				Je																			
ARACEAE																									
Acorus calamus*						F																			
Colocasia antiquorum*						Fc																			
Stylochiton natalense										J												V			
Zantedeschia albomaculata*				D									Nr												
ARECACEAE																									
Phoenix reclinata										Jr															
COMMELINACEAE																									
Commelina africana		B											M	N						T					
Cyanotis speciosa	A																								
CYPERACEAE																									
Cyperus esculentus				D		F																			
Cyperus longus*						Fi					Ks											Vs			

Cyperus sexangularis Fi
Cyperus sp. B F

DIOSCOREACEAE
Dioscorea diversifolia N
Dioscorea dregeana Dp N T
Dioscorea dumentorum* Y
Dioscorea rupicola Y
Dioscorea sylvatica* Y

HYPOXIDACEAE
Empodium plicatum J N
Hypoxis gerrardii Fd
Hypoxis latifolia* AeB G MeNe
Hypoxis rooperi L
Hypoxis sp. Fa N T

IRIDACEAE
Aristea cognata U
Aristea ecklonii Jc Kc
Aristea woodii I
Belamcanda chinensis N
Belamcanda sp. F
Crocosmia aurea Fd
Crocosmia paniculata B Fdc
Crocosmia pottsii B Fdc
Dierama pendulum I
Dietes iridioides D Fdc Y
Gladiolus aurantiacus Ke
Gladiolus dalenii* Ae Fd Ks Ne
Gladiolus papilio Ne
Gladiolus sericeo-villosus Bc Dc Fdc U I
Gladiolus spp. E I
Hesperantha baurii Je I
Hesperantha lactea F
Hesperantha sp. F
Moraea sp. Fdc
Moraea spathulata* B N
Sparaxis grandiflora N I
Tritonia disticha E
Tritonia lineata ErFi
Tritonia sp. I
Watsonia densiflora I Y

LILIACEAE
Agapanthus africanus* Ae DpE Je MeNe I
Agapanthus praecox A D I
Albuca spp.* C Fa I
Aloe arborescens D I Y
Aloe candelabrum E Fi
Aloe cooperi Dp Y
Aloe ferox T V
Aloe gracilis I
Aloe maculata Fc Kic Y
Aloe marlothii E Fia
Aloe sessiliflora E Fi
Aloe sp. B D V
Aloe spectabilis E Fi
Aloe tenuior Fa I
Aloe thraskii E Fi
Behnia reticulata Ae
Bowiea volubilis* Ae Dp F G L T V I
Bulbine alooides C O
Bulbine frutescens Ne
Bulbine latifolia Cc Fd6 N T
Bulbine sp. F
Chlorophytum modestum* I
Dipcadi brevifolium* M I
Dipcadi sp. I
Dipcadi viride C I
Drimia robusta* Kc
Drimiopsis maculata Fi
Eriospermum mackenii I
Eriospermum sp.* B Fe I
Eucomis autumnalis* C Dp Fi5 JeKec Oc I
Eucomis comosa O Qic
Eucomis sp. F O Q
Gasteria croucheri N I
Gloriosa superba* AeB Ne W
Haworthia limifolia F I
Kniphofia buchananii I
Kniphofia laxiflora J
Kniphofia parviflora I
Kniphofia rooperi J I
Kniphofia uvaria Dc

Ledebouria cooperi Dp Fi N Y
Ledebouria ovatifolia* Dc Fdc K O
Littonia modesta* Ae Ne
Myrsiphyllum asparagoides Ae V
Ornithogalum longibracteatum* I
Protasparagus sp. J
Protasparagus virgatus Ae C Fa
Sandersonia aurantiaca* Ae I
Sansevieria hyacinthoides Fa N V I
Scilla natalensis* Dp Fi Ic O R U Y
Scilla nervosa* Fi K Y
Smilax kraussiana Vs
Trachyandra gerrardii Er Ti
Tulbaghia alliacea A F K O I
Tulbaghia natalensis I
Tulbaghia violacea Ae Fc L I
Urginea altissima* Fc J Me V I
Urginea delagoensis* I
Urginea macrocentra* Fa
Urginea physodes* Dp Y

ORCHIDACEAE
Acrolophia cochlearis Ae
Ansellia gigantea Ae Ns I
Brachycorythis ovata N I
Brachycorythis sp. I
Corycium nigrescens Ne
Cyrtorchis arcuata Ae
Disa aconitoides Be
Disa polygonoides N
Disa stachyoides I
Disa versicolor I
Eulophia angolensis Ae
Eulophia clavicornis 1 B Dp I
Eulophia clavicornis 2 I
Eulophia clavicornis 3 B I
Eulophia cucullalata AeB
Eulophia ensata Ae
Eulophia leontoglossa I
Eulophia ovalis AeB Or I
Eulophia petersii I
Eulophia sp. Ne I
Eulophia speciosa I
Eulophia tenella B Dp Nc I
Eulophia welwitschii Ae
Habenaria dives R
Habenaria dregeana I
Habenaria epipactidea Fe Nr I
Habenaria sp. B
Mystacidium millarii Ae
Mystacidium venosum Ae
Oeceoclades mackenii Ae
Polystachya ottoniana A
Satyrium bracteatum Fa
Satyrium longicauda I
Satyrium parviflorum Ae I

POACEAE
Coix lacryma-jobi Q1 T I
Cymbopogon marginatus Fa1e J I Y
Cymbopogon excavatus F
Cymbopogon validus Fe K
Eragrostis plana B D H U
Imperata cylindrica F K
Setaria megaphylla U

TYPHACEAE
Typha capensis C Dp Y

ZINGIBERACEAE
Kaempferia sp. J N I
Siphonochilus aethiopica Y
Siphonochilus natalensis J K

ANGIOSPERMAE - DICOTYLEDONAE

ACANTHACEAE
Adhatoda densiflora Fe
Barleria ovata Ae Hc
Blepharis capensis* G Sr
Crabbea hirsuta I
Crabbea nana Q S I
Hypoestes aristata V Y
Justicia flava J

Peristrophe hensii V
Thunbergia atriplicifolia E T
Thunbergia dregeana A T
Thunbergia natalensis Ae
Thunbergia venosa Ae N

AIZOACEAE
Aizoon canariense Fe
Corbichonia decumbens Fe

AMARANTHACEAE
Achyranthes aspera Fe Me
Achyropsis avicularis Fe Ke T
Achyropsis leptostachya Fe Ke
Herbstaedtia odorata F
Pupalia atropurpurea Ke
Pupalia sp. B

ANACARDIACEAE
Harpephyllum caffrum Fe He
Ozoroa paniculosa* Fcd Jc Y
Protorhus longifolia* M Or R T
Rhus chiridensis M
Rhus laevigata Fi
Sclerocarya birrea* F He N X

ANNONACEAE
Annona senegalensis Fe N
Monanthotaxis caffra Nse X

APIACEAE
Alepidea amatymbica F JcKs O X
Alepidea longifolia J
Anethum graveolens N R X
Berula erecta* K
Bupleurum mundtii Ae
Foeniculum vulgare Ae N R X
Heteromorpha trifoliata* Fad I J Ls N X Y
Lichensteinia interrupta* Fc J K N R X
Lichensteinia kolbeana Fc J
Pimpinella caffra Ae Fa X
Pimpinella sp. Ae

APOCYNACEAE
Acokanthera oblongifolia* Fa SeT Y
Acokanthera oppositifolia* Fa6 H Sesr Y
Catharanthus roseus G H I T
Oncinotis inandensis Ae N
Rauvolfia caffra* Dp F G H I Ke X
Strophanthus gerrardii* X
Strophanthus petersianus* X
Strophanthus speciosus* Ss Y
Wrightia natalensis E

AQUIFOLIACEAE
Ilex mitis Fi Ke T X

ARALIACEAE
Cussonia spicata C Fe Ke

ASCLEPIADACEAE
Asclepias cucullata X
Asclepias cultriformis X
Asclepias flexuosa X
Asclepias fruticosa* Dp Fi Je L
Asclepias gibba G
Asclepias meliodora X
Asclepias physocarpa* Ec X
Asclepias sp.* B Y
Ceropegia woodii Ne
Cynanchum obtusifolium* Ne X
Huernia hystrix X
Huernia sp. X
Orbea variegata X
Pachycarpus inconstans F Ls Ns X Y
Pachycarpus natalensis F Y
Pachycarpus scaber X Y
Sarcostemma viminale* D E Fe V
Schizoglossum atropurpureum X
Secamone gerrardii Fc M Or U
Secamone sp. Ae
Sisyranthus huttoniae N
Sisyranthus saundersiae Ne
Sisyranthus trichostomus Ne
Stapelia gigantea NeOr U X
Tylophora flanaganii F
Xysmalobium involucratum X
Xysmalobium sp. Y
Xysmalobium undulatum F H Ls Ne T X Y

ASTERACEAE
Achyrocline stenoptera Ns
Adenostemma caffrum Ae Fi K
Artemisia afra F H Ks X
Aspilia natalensis Fde JeKc T
Aster bakeranus* C Fi6 J K L N S T Vs X
Aster serrulatus F N
Aster spp. L Se
Athrixia phylicoides F H J
Berkheya rhapontica J
Berkheya seminivea Fe
Berkheya sp. G Hc T U V
Berkheya speciosa Fc6 V
Berkheya subulata Fi V
Bidens pilosa Fcdr Or
Blumea alata F
Brachylaena discolor Fa H J N
Brachylaena elliptica Dp Fe Je N O Q Y
Callilepis laureola* B Dp Fa Ni S Y
Conyza podocephala J K T
Conyza scabrida JrKc R Ur X
Conyza ulmifolia J
Cotula anthemoides Fi K
Crassocephalum rubens Fe
Dicoma anomala D Fad Hie J Q T W
Dicoma sp. J Nr
Dicoma speciosa J
Dicoma zeyheri J
Eclipta prostrata* Ae R
Ethulia conyzoides Fa
Felicia erigeroides Fac
Gazania rigens Ae
Gerbera ambigua Fa J
Gerbera piloselloides Fa J Ks V
Helichrysum cooperi X
Helichrysum decorum Ns
Helichrysum foetidum D T
Helichrysum nudifolium Fi JeKs Ns T U
Helichrysum oxyphyllum X
Helichrysum pedunculatum J K T
Helichrysum pilosellum Fi
Helichrysum spp. D F T
Matricaria nigellifolia* K T Y
Melanthera scandens Fe
Microglossa aespilifolia I K Y
Mikania capensis C G K Ls T Y
Mikania natalensis C Ls T Y
Nidorella sp. S U
Osteospermum grandidentatum Fe
Osteospermum imbricatum Fe Ke
Osteospermum jucundum F
Osteospermum scariosum Fe
Othonna natalensis Fa H Y
Pentzia pilulifera T
Phymaspermum acerosum X
Platycarpha glomerata X
Printzia pyrifolia* D F Ic N Y
Pulicaria scabra D H I K V
Schistostephium hippiifolium k
Senecio bupleuroides* Dp J M
Senecio erubescens Fc J k
Senecio inornatus J M
Senecio lanceus Dp
Senecio oxyriifolius B K Kr Y
Senecio quinquelobus F J Kc X
Senecio serratuloides* H J G T
Senecio sp. C Jce
Senecio speciosus Gr Jr Ls U
Senecio tamoides Fc Y
Spilanthes mauritiana Q
Ursinia anthemoides
Ursinia sp. V
Ursinia tenuiloba J
Vernonia adoensis F J Kcr U
Vernonia angulifolia Fc
Vernonia hirsuta F J Ls
Vernonia mespilifolia Ae K
Vernonia natalensis X
Vernonia neocorymbosa Dp F N Y

Vernonia oligocephala Fc X
Vernonia sp. F

BALANITACEAE
Balanites maughamii N X

BALANOPHORACEAE
Sarcophyte sanguinea Fd I

BALSAMINACEAE
Impatiens hochstetteri T

BEGONIACEAE
Begonia sp. Fc Je X
Begonia sutherlandii Fe X

BIGNONIACEAE
Kigelia africana* F X

BORAGINACEAE
Cynoglossum hispidum Ae F
Myosotis afropalustris N

BRASSICACEAE
Heliophila subulata X
Lepidium capense J
Lepidium pinnatum J
Lepidium schinzii C J
Lepidium sp. C

CACTACEAE
Rhipsalis baccifera Je

CAMPANULACEAE
Wahlenbergia grandiflora N Y
Wahlenbergia sp. Fi
Wahlenbergia undulata Fe V X

CANELLACEAE
Warburgia salutaris FcG JsKec T U

CANNABACEAE
Cannabis sativa DsE J Y

CAPPARACEAE
Cadaba natalensis Je Ne R
Capparis brassii Ae Je
Capparis tomentosa* B H I Jes Lr N S U X Y
Cleome monophylla Ae

CELASTRACEAE
Cassine aethiopica Fdc Y
Cassine crocea* Fe
Cassine sp. Ji
Cassine transvaalensis Fce K
Catha edulis* F
Maytenus heterophylla* Fde Jr S U Y
Maytenus spp. Fc J S
Pterocelastrus rostratus Or
Salacia leptoclada A

CHENOPODIACEAE
Chenopodium album* Fa T W Y
Chenopodium ambrosioides* F J W Y

CLUSIACEAE
Hypericum aethiopicum* C D F Gc T V

COMBRETACEAE
Combretum apiculatum Fc
Combretum caffrum X
Combretum erythrophyllum* Y
Combretum hereroense Fc
Terminalia phanerophlebia Jr Ne R
Terminalia sericea* Jr Ne R

CONVOLVULACEAE
Convolvulus farinosus Fe
Ipomea ficifolia F S
Ipomoea calrica* F K Ne
Ipomoea congesta F
Ipomoea crassipes Fds Y
Ipomoea purpurea C F He
Merremia tridentata F

Turbina oblongata X

CRASSULACEAE
Cotyledon orbiculata* Cc Q T U V X
Crassula alba Fde Ks Ne
Crassula barbata Fe
Crassula dependens X
Crassula multicava Fe
Crassula orbicularis Fe
Crassula sarcocaulis Fe
Crassula sp. X
Crassula vaginata Ae U X
Kalanchoe crenata U X
Kalanchoe nr. paniculata X
Kalanchoe rotundifolia* Fe X
Kalanchoe sp. V X
Kalanchoe thyrsiflora* Dp Fac V X

CUCURBITACEAE
Coccinia palmata Fc
Cucumis africanus* NeOc Y
Cucumis hirsutus* F Jc
Cucumis myriocarpus* A T X
Cucumis sp.* F
Gerrardanthus macrorhizus F
Lagenaria sphaerica* Fc H I
Momordica balsamina* F T
Momordica foetida F H

CUNONIACEAE
Cunonia capensis N

DIPSACAEAE
Cephalaria attenuata Ki N X
Cephalaria zeyheriana H I O
Scabiosa columbaria Ae D T V

EBENACEAE
Diospyros pallens Fd
Diospyros villosa Fc O Ur
Diospyros whyteana* Dc
Euclea crispa* D Fc J N O X
Euclea divinorum F Q
Euclea natalensis* C FcGcH IsJr
Euclea schimperi D
Euclea sp. C Je
Euclea undulata Dp Fc L

EUPHORBIACEAE
Acalypha peduncularis H Je
Acalypha punctata Je
Acalypha sp. Ls
Andrachne ovalis* Je Ls Se M I
Andrachne sp. S
Antidesma venosum* Fcd
Bridelia cathartica Ae N R
Bridelia micrantha* Fe Jr
Cavacoa aurea K O Vs
Clutia affinis F
Clutia heterophylla Y
Clutia hirsuta Ae Fc H K X Y
Clutia platyphylla F
Clutia pulchella* E Fdı L Ur I Y
Clutia sp.* O S
Croton gratissimus* Ae D F Gr IrJrKce N Q U
Croton sylvaticus* A D F GrH IrJrKce U
Euphorbia bupleurifolia Fe Ir T
Euphorbia ingens* F I X
Euphorbia nr. woodii D T
Euphorbia pugniformis Ir T
Jatropha hirsuta Kc T X
Jatropha zeyheri H T
Margaritaria discoidea* Ae
Monadenium lugardae* F N
Phyllanthus meyerianus J K L
Phyllanthus reticulatus N X
Ricinus communis* Fc Q T Y
Spirostachys africana* F Ls Ti Vs Y
Synadenium cupulare* J Ls Q R X
Tragia meyeriana G

FABACEAE
Abrus precatorius* Ae J
Acacia caffra Ae Fi He

Acacia gerrardii Nces
Acacia karroo Fdi Ne
Acacia nilotica J
Acacia rehmanniana Oc T
Acacia xanthophloea Ke
Afzelia quanzensis T
Albizia adianthifolia* Ae F T V
Albizia forbesii N
Albizia sp. Kc
Albizia suluensis* Kc N
Alysicarpus rugosus JeKe
Argyrolobium sp. T
Argyrolobium marginatum F
Argyrolobium natalense Ae
Argyrolobium speciosum Be
Argyrolobium stipulaceum S
Argyrolobium tomentosum Ne
Calpurnia aurea T W Y
Calpurnia villosa W Y
Cassia floribunda I
Cassia italica* J K
Cassia mimosoides Fd N T
Cassia occidentalis* F N O S W I
Cassia sp. I
Crotalaria lanceolata* I
Crotalaria sp.* J
Dalbergia obovata Ae Qi S
Dialium schlechteri T
Dichrostachys cinerea Or
Elephantorrhiza elephantina* Ae C Fdce J Ke T
Elephantorrhiza sp.* Mer
Entada spicata Cr MeNe T Y
Eriosema cordatum A B Y
Eriosema distinctum B G
Eriosema salignum B Y
Erythrina caffra* C G V
Erythrophleum lasianthum* F KsLs N R Y
Indigofera cylindrica Fa Y
Indigofera eriocarpa Fd
Indigofera hilaris* Fd
Indigofera micrantha Fd
Indigofera spp. Fd J K N I
Indigofera tenuissima* J
Indigofera velutina Fd
Lonchocarpus capassa* Fd
Lotus discolor J
Macrotyloma axillare Y
Millettia grandis Ns
Mundulea sericea* Ne
Newtonia hildebrandtii N I
Ormocarpum trichocarpum Ne
Peltophorum africanum Bc
Psoralea pinnata Ne
Rhynchosia nervosa Ae
Rhynchosia sp. A L
Schotia brachypetala Fd He
Sphenostylis angustifolia Fe He
Tephrosia capensis* Fe LsM N
Tephrosia densiflora* W
Tephrosia diffusa* T W
Tephrosia elongata Fe
Tephrosia grandiflora Je W
Tephrosia kraussiana J
Tephrosia lurida Ae I
Tephrosia macropoda* J K T W Y
Tephrosia purpurea* W
Tephrosia toxicaria* W
Tephrosia vogelii W
Vigna luteola Ae Ke
Vigna unguiculata Ae Ke
Vigna vexillata Ae Ke
Zornia capensis I

FLACOURTIACEAE
Dovyalis rhamnoides K
Gerrardina foliosa Dp K L N I
Homalium dentatum F
Trimeria grandifolia Fc
Xylotheca kraussiana Ae

GENTIANACEAE
Chironia purpurascens F
Sebaea bojeri Ae
Sebaea crassulifolia Fd S
Sebaea grandis Ae
Sebaea sedoides Ae Fi
Sebaea sp. Ae

GERANIACEAE
Geranium ornithopodon JsKi
Monsonia emarginata Fd S Y
Pelargonium luridum Fd I Y
Pelargonium sidifolium C Fd
Pelargonium spp. Fd J K I Y

GESNERIACEAE
Streptocarpus prolixus Y
Streptocarpus sp. D

GREYIACEAE
Greyia sutherlandii Fe

HALORAGACEAE
Gunnera perpensa B Dp G K Y

ICACINACEAE
Apodytes dimidiata Fa N I Y
Pyrenacantha grandiflora Ae
Pyrenacantha scandens A B Dp

ILLECEBRACEAE
Dianthus crenatus Ae Ne I
Pollichia campestris Js
Silene bellidioides Ae
Silene burchellii Ae

LAMIACEAE
Becium kwanum T
Becium obovatum Fi T
Leonotis leonurus Fd JcK Ls Se Y
Leonotis ocymifolia F K Y
Leucas martinicensis Ki
Mentha aquatica K I
Mentha longifolia J KcL
Plectranthus ambiguus K
Plectranthus hadiensis Fc W Y
Plectranthus laxiflorus Fc Kc Q Y
Plectranthus madagascariensis Fc J T
Plectranthus sp. K
Plectranthus sp. K
Plectranthus sp. J
Pycnostachys reticulata Q
Salvia scabra E
Stachys nigricans Ae
Syncolostemon densiflorus Ae
Syncolostemon parviflorus Fi He
Tetradenia riparia F G JeK Y
Teucrium kraussii H Se

LAURACEAE
Cryptocarya latifolia J
Ocotea bullata G Ls Or U

LECYTHIDACEAE
Barringtonia racemosa* Ke

LENTIBULARIACEAE
Utricularia livida Fi
Utricularia prehensilis Ae

LINACEAE
Linum thunbergii He O S I Y

LOBELIACEAE
Cyphia elata Fe
Lobelia coronopifolia D
Lobelia erinus Ks I
Lobelia patula Qi
Monopsis decipiens K
Monopsis scabra K

LOGANIACEAE
Buddleia salviifolia* R I
Nuxia floribunda N
Nuxia oppositifolia A
Strychnos decussata F
Strychnos henningsii* D Fa K O
Strychnos spinosa Ke Se V

LORANTHACEAE
Erianthemum dregei Fi Y

MALPIGHIACEAE
Acridocarpus natalitius N I

MALVACEAE
Hibiscus aethiopicus F U
Hibiscus pedunculatus G
Hibiscus pusillus Ae Ne
Hibiscus surattensis C G U
Sida dregei T
Thespesia acutiloba I

MELASTOMATACEAE
Dissotis canescens Fd

MELIACEAE
Ekebergia capensis Ae Fae H J Ne
Trichilia dregeana Fcd O
Trichilia emetica* Fd6c Oc
Turraea floribunda* Ge MeNeOe
Turraea obtusifolia* Fce

MELIANTHACEAE
Bersama lucens* B D P I
Melianthus comosus* F H O S T U V
Melianthus dregeanus* F H S

MENISPERMACEAE
Cissampelos mucronata I
Cissampelos sp. Dp
Cissampelos torulosa C Ic Q
Stephania abyssinica H I

MESEMBRYANTHEMACEAE
Aptenia cordifolia Ae Fic Nr T
Carpobrotus edulis Fi6 J
Delosperma velutinum Fe
Mesembryanthemum sp. Fce Ir NeOr U

MORACEAE
Ficus ingens H Y
Ficus natalensis Dp
Ficus sur J Y
Ficus sp. H

MYRICACEAE
Myrica serrata D J K L I

MYRSINACEAE
Embelia ruminata* Fa H
Maesa alnifolia Ae Fa
Maesa lanceolata* Fae
Maesa sp. Fa
Rapanea melanophloeos Fec J M

MYRTACEAE
Eucalyptus sp. Ks M
Eugenia albanensis Fd
Heteropyxis natalensis Y
Syzygium cordatum* Fe J
Syzygium gerrardii J

OCHNACEAE
Ochna natalitia B
Ochna serrulata Fc

OLEACEAE
Jasminum multipartitum Ae
Jasminum sp.* Fc
Olea europaea G J L I

OXALIDACEAE
Oxalis semiloba* Qi

PASSIFLORACEAE
Adenia gummifera* Fe H Ke M P I
Basananthe sandersonii A

PEDALIACEAE
Ceratotheca triloba D F

PERIPLOCACEAE
Mondia whitei F H
Raphionacme sp. Fa H I K
Tacazzea apiculata H

PHYTOLACCACEAE
Phytolacca americana* Fe I J S T M
Phytolacca dodecandra* Bc G Ke
Phytolacca heptandra* F Gc J Ne S Y
Phytolacca octandra* J Ne S T Y

PITTOSPORACEAE
Pittosporum viridiflorum Fce H Ke N Oc

PLANTAGINACEAE
Plantago major DpE Fd T

PLUMBAGINACEAE
Plumbago auriculata* Ls Ne Ur I

POLYGALACEAE
Polygala fruticosa Dp G H IsJ
Polygala galpinii Ae
Polygala hottentotta Fc I Y
Polygala serpentaria Ae Fi
Polygala sp. J
Polygala virgata He

POLYGONACEAE
Emex australis Fc Y
Oxygonum dregeanum Fe
Polygonum lapathifolium C
Polygonum salicifolium T
Rumex lanceolatus B D FaidH T Y
Rumex sagittatus F J K Ls O Q

PORTULACACEAE
Portulaca quadrifida Fe
Talinum caffrum Fci Je N

PTAEROXYLACEAE
Ptaeroxylum obliquum* Es Ls I Y

RANUNCULACEAE
Anemone caffra Fce N R I
Clematis brachiata C Fac K Ls T I
Knowltonia anemonoides Cc Ls
Ranunculus multifidus* C E Fd G I J Ls T
Ranunculus sp. Ls I Y
Thalictrum rhynchocarpum I

RHAMNACEAE
Berchemia discolor T I
Berchemia zeyheri Fdi Oc I
Helinus integrifolius Ae Ne
Rhamnus prinoides H J U I
Ziziphus mucronata I Je O Q T U

ROSACEAE
Agrimonia eupatoria Fa J
Leucosidea sericea Fa V I
Prunus africana* R I
Rubus ludwigii F O
Rubus pinnatus J V
Rubus rigidus Fd O V

RUBIACEAE
Agathisanthemum bojeri Ae
Burchellia bubalina Ae Ne I
Canthium ciliatum Fa N
Canthium inerme Fd
Catunaregam spinosa* Ke
Coddia rudis B Ke
Conostomium natalense Ae
Gardenia cornuta Fe
Gardenia ternifolia Ke
Gardenia thunbergia Fe Ke
Gardenia volkensii Fe
Hyperacanthus amoenus Fe
Kohautia amatymbica AeB E Hi Ne
Lagynias lasiantha Fd
Oldenlandia affinis J M
Oldenlandia corymbosa Ae N I
Oldenlandia sp. Ae
Pavetta spp. H N

Pentanisia prunelloides C Dp Fc H I J K N O S T U
Pentodon pentandrus Fe
Psychotria capensis Fe
Rubia cordifolia AeB D F J N Q X
Spermacoce natalensis Fd J M N T

RUTACEAE
Calodendrum capense X
Clausena anisata EsFac Ki MsNeO
Vepris undulata F Ke
Zanthoxylum capense B Fac H IsJe1 Or Q S T
Zanthoxylum davyi H J K Ser Y

SALVADORACEAE
Azima tetracantha Q

SANTALACEAE
Osyridicarpos schimperianus X

SAPINDACEAE
Cardiospermum halicacabum* C Fdc H Js T
Deinbollia oblongifolia Fd N
Hippobromus pauciflorus* Ae Fd Ls Ne V Y

SAPOTACEAE
Sideroxylon inerme Kc Ne Ur Y

SCROPHULARIACEAE
Buchnera dura Ae X
Cycnium racemosum D Oe
Cycnium tubulosum Ae X
Graderia scabra Aes D Fc K T X
Halleria lucida V X
Harveya speciosa F N U
Manulea parviflora Fi
Peliostomum calycinum Fi K T
Sutera floribunda D J
Sutera kraussiana D
Sutera sp. Ke Ns
Zaluzianskya maritima Ae

SELAGINACEAE
Hebenstretia sp. X
Selago sp. Sr U
Tetraselago natalensis C

SOLANACEAE
Datura aetel* Ls N T
Datura stramonium* I Ls N T U
Physalis peruviana* Fi
Solanum sp.* B F
Solanum acanthoideum T
Solanum aculeastrum* Or Q Ur Y
Solanum aculeatissimum* S
Solanum capense C G Q S T Vr
Solanum giganteum J
Solanum incanum* Q T
Solanum mauritianum* Fe
Solanum nigrum* Fi Or T
Solanum panduriforme Q T
Solanum tomentosum C Q T

Withania somnifera* C Fi I J Ki Nc T Y

STERCULIACEAE
Dombeya rotundifolia Fc M
Hermannia depressa Ae Fd J N X
Hermannia grandistipula Y

THYMELAEACEAE
Gnidia anthylloides* J Ks Ns S
Gnidia calocephala T
Gnidia cuneata H Kc Q S T
Gnidia kraussiana* Dp Fc J O S Y
Gnidia polyantha Ae T
Gnidia spp. H S T V

TILIACEAE
Corchorus asplenifolius A
Grewia caffra Gc
Grewia occidentalis D Gc T X
Triumfetta pilosa T
Triumfetta rhomboidea Dp

TRIMENIACEAE
Xymalos monospora F

ULMACEAE
Chaetacme aristata* F Q
Trema orientalis Fe

URTICACEAE
Laportea grossa Y
Laportea peduncularis Y
Pouzolzia mixta Kc X
Urtica urens B Y

VERBENACEAE
Clerodendrum glabrum FaGe J K Y
Clerodendrum hirsutum Fa I
Clerodendrum myricoides S
Clerodendrum suffruticosum S
Clerodendrum triphyllum Dp Gc I
Lantana rugosa Fi J T V
Lippia javanica F J K T
Priva cordifolia T V
Vitex rehmannii Fc J N
Vitex wilmsii J NeGc

VIOLACEAE
Hybanthus enneaspermus H
Hybanthus sp. Ae

VISCACEAE
Viscum sp. Fi J N Q T

VITACEAE
Cissus quadrangularis Y
Cyphostemma hypoleucum Kc
Cyphostemma lanigerum Q
Cyphostemma natalitium Dp Y
Rhoicissus tomentosa* Y
Rhoicissus tridentata* B Dp K X